Hands-on Probabilistic Machine Learning: Bayesian Model Updating and Gaussian Processes

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September 2026

Learning objectives of this lecture

L2: Hands-on Probabilistic Machine Learning

- Bayesian Model updating – using MCMC
- Gaussian Processes for Regression
- A hands-on case study with real aeroacoustic wind-tunnel data (NASA airfoil self-noise): system identification via probabilistic machine learning

L1: Fundamentals of Probabilistic Machine Learning (PML) → L2: Hands-on PML → L3: Fundamentals of Physics-Enhanced Machine Learning (PEML) → L4: Hands-on PEML

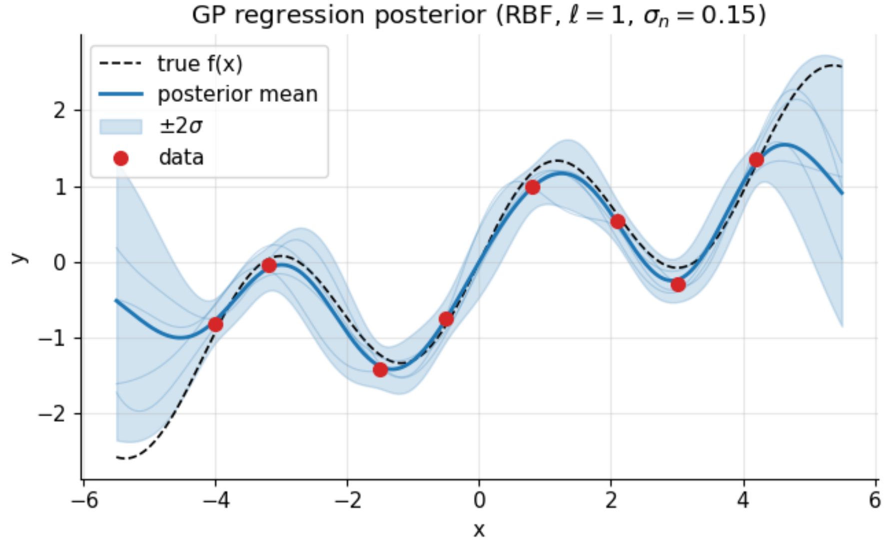
For physics-based models: uncertainty in **model parameters (repeat from L1)**

Can be handled by

- **Bayesian inference**
  - yields a posterior distribution over the **latent parameters** given the data, a physics-based forward model, an error (likelihood) model and a prior.
  - by using sampling (e.g. Markov-chain Monte Carlo) or variational inference

$$\text{Posterior} = \frac{\text{Likelihood} \cdot \text{Prior}}{\text{Evidence}}$$

- **Gaussian processes (GPs)**
  - offer a complementary, non-parametric Bayesian route:
  - a GP places a distribution over functions and returns a predictive mean with a variance that grows where data are sparse, so the variance encodes epistemic uncertainty about the response;
  - used as an emulator or as an additive discrepancy term within Bayesian calibration,
  - a GP also supports inference about **fixed-but-unknown physical parameters**



Next Lecture

Bayesian Model Updating: obtaining the posterior distribution on the physics-based model parameters of interest

Prior knowledge on uncertain input parameters

Prior distribution

prior knowledge, *experts' opinion, physics constraints, and prior data on uncertain input model parameters that cannot be directly observed or measured (latent)*

Likelihood

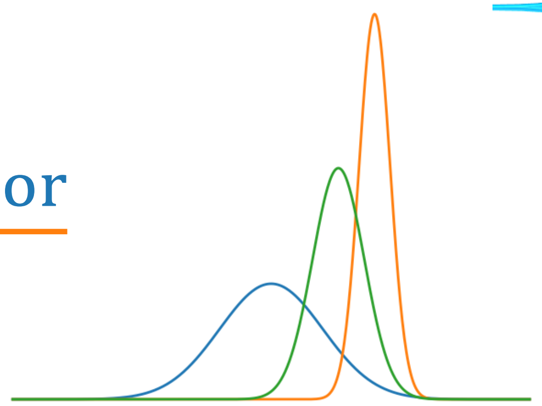
Discrepancy between the observed informative data and model prediction is described with *a statistical model*

Known physics-based model

Noisy dynamics response measurements

Learn

$$\text{Posterior} = \frac{\text{Likelihood} \cdot \text{Prior}}{\text{Evidence}}$$



Bayesian model updating results are sensitive to:

- Prior distribution assumption: **uncertainty on prior:** limited prior knowledge (e.g. new manufacturing process) and/or conflicting experts' opinion
- Likelihood assumptions
- (informative) measurements can be **expensive to acquire** and/or **difficult to obtain → limited & noisy**
- **computational strategy to obtain the posterior**

Fundamentals of Bayesian Model updating: all done in the notebook (about 30 minutes):

L2 Part A: Bayesian Model Updating

The Kennedy & O'Hagan Framework for Engineers

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file name: L2_fileA_Bayesian_Model_Updating_Kennedy_OHagan.ipynb

Part of Lecture 2 - first notebook

Instructor: Alice Cicirello

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The first draft version of the notebook was generated using Anthropic's Claude

Date: September 2026

Last Modifications and checks made by Alice Cicirello on: 5/09/2026

Note: we are going to investigate only a standard MCMC sampling strategy.
However, there are many SOTA efficient sampling strategies (e.g. SMC, TMCMC), and alternative to sampling (e.g. Variational Inference)

Gaussian Processes: theory and practice
all done in the notebook (about 30 minutes):

L2: Gaussian Processes: theory and practice

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file name: L2_fileB_gaussian_processes_intro.ipynb

Part of Lecture 2 - second notebook

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Introduction

This notebook develops Gaussian Process (GP) regression *from the probability up*.

Every equation is paired with a small, runnable numerical experiment so you can see the maths in action.

By the end you should be able to

(i) explain what a GP *is*,

Your turn to apply these concepts to a new problem: **System Identification** (~30 mins) - building **mathematical models of physical systems from measured input–output data** - not just learning the parameters of a model!

Hands-on case study with real aeroacoustic wind-tunnel data (NASA airfoil self-noise)

Probabilistic Machine Learning for System Identification

A hands-on case study with real aeroacoustic wind-tunnel data (NASA airfoil self-noise)

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file name: L2_hands_on_1_probabilistic_ML_sysID_aeroacoustics.ipynb

Part of Lecture 2 - third notebook (comparison of results obtained with Bayesian inference and Gaussian Processes)

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Why should *you* care?

Every fan, compressor, propeller and wind-turbine blade you will ever design radiates noise, and much of it comes from the blade sections themselves (*self-noise*). Certification limits are written in decibels, wind-tunnel time costs thousands of pounds per hour, and the semi-empirical models you will meet in industry (e.g. the Brooks–Pope–Marcolini "BPM" model used inside wind-turbine noise codes) were themselves *identified from data*.

During this lecture you learned about

Bayesian Model updating – using MCMC
Gaussian Process for Regression
System identification via probabilistic machine learning

Next Lecture

- Fundamentals of Physics-Enhanced Machine Learning

Further readings

- **Deep Learning: foundations and concepts, C. Bishop, H. Bishop** (https://www.bishopbook.com/)
- **Pattern Recognition and machine learning, C. Bishop** (https://www.microsoft.com/en-us/research/uploads/prod/2006/01/Bishop-Pattern-Recognition-and-Machine-Learning-2006.pdf)
- **Deep Learning, I. Goodfellow, Y. Bengio, A. Courville** (https://www.deeplearningbook.org/)

I attempted to credit all the content/figures where I could find the original source. Sometimes the material is so widely used that it's difficult to be sure of the original author. Please let me know if you find missing references.